

Vigneshwari Jayaprakash

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Data Scientist who builds ML systems that ship and scale. End-to-end experience spanning predictive modeling, NLP, and LLM integration — delivering \$1.4M+ in documented impact across finance and sourcing operations at BNSF Railway, from first commit to production.

TECHNICAL SKILLS

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, SciPy), SQL, PySpark, R
ML & Statistics: XGBoost, Random Forest, Logistic Regression, LSTM, PyTorch, TensorFlow, Hypothesis Testing, A/B Testing, Causal Inference, Feature Engineering, SHAP
NLP & GenAI: LangChain, LangGraph, RAG Systems, ChromaDB, Transformers, Prompt Engineering, LLM Fine-Tuning
MLOps & Infra: MLflow, FastAPI, Docker, AWS (S3, Lambda), Databricks, Snowflake, CI/CD, GitLab, Evidently.ai
Analytics & Visualization: Tableau, Power BI, Streamlit, Exploratory Data Analysis (EDA), Executive Reporting

EDUCATION

Arizona State University Tempe, AZ
M.S. in Data Science (Computing & Decision Analytics) — GPA: 4.0/4.0 | ASU Scholar Aug 2024 – May 2026
Coursework: Machine Learning, Statistical Modeling, Data Mining, Distributed Systems, Data Visualization
Anna University India
B.Tech in Information Technology — GPA: 3.6/4.0 | Gold Medalist Jun 2009 – Apr 2013

PROFESSIONAL EXPERIENCE

New Mexico Department of Information Technology NM, USA
Data Scientist — AI Systems & Anomaly Detection Jun 2025 – Aug 2025

- Deployed XGBoost + Isolation Forest on AWS via FastAPI, hitting 85% detection accuracy and 60% fewer false positives on 10K+ incidents.
- Conducted EDA and feature engineering on incident datasets, surfacing predictive signals that improved F1 score 22%.
- Built a LangGraph + RAG platform with ChromaDB search, cutting analyst triage time 70% across engineering and operations teams.
- Optimized batched inference with semantic caching, reducing prediction latency 40% on AWS.

Infosys Ltd (Client: BNSF Railway) India
Technology Analyst — Data Science & ML Engineering Oct 2013 – Jan 2019

- Generated \$1M+ annual savings owning fraud detection on 1M+ daily transactions at sub-200ms latency, cited by CFO.
- Lifted F1 score 18% via 5-round A/B testing, experiment design, and causal inference; cut sign-off cycles 30% with SHAP-driven Power BI dashboards.
- Built customer segmentation models using SQL window functions and K-means clustering, enabling targeted risk profiling across 500K+ daily records.
- Saved \$400K/year deploying time-series regression models for 500+ assets via MLflow + PySpark on Databricks at 99.9% uptime.
- Standardized ML model deployment across dev/test/prod using Docker and CI/CD, compressing release cycles 3×.
- Mentored 5 junior analysts in Tableau and data storytelling, reducing dashboard turnaround time 35% and improving report adoption across business stakeholders.
- Promoted 3 times in 5 years: Software Engineer Trainee | Software Engineer | Senior Software Engineer | Technology Analyst.

PROJECTS

Real-Time Multi-Domain Risk Intelligence Platform | *Python, XGBoost, LSTM, PyTorch, LangChain, FastAPI, Streamlit*

- Built a real-time risk platform — fraud classification on 590K transactions (94% recall) and LSTM time-series model predicting equipment failure 72 hours ahead on NASA sensor data.
- Integrated LangChain LLM explanations and Evidently.ai drift monitoring, deployed on AWS via FastAPI and Docker CI/CD.
- Reduced manual risk review 40% via a Streamlit + SHAP dashboard enabling data-driven decisions for operations and risk teams.

SentinelRAG — NLP Compliance Screening & Audit System | *Python, LangChain, ChromaDB, FastAPI, Docker, AWS*

- Engineered a hybrid NLP/RAG screening system processing 18,700+ entities with 99% recall, 100% precision, and full deterministic audit trail.
- Deployed on AWS via FastAPI with Docker CI/CD, ChromaDB transformer embeddings, and JSON audit log per entity.
- Cut query latency 40% via batched inference and semantic caching; results surfaced through a Streamlit compliance dashboard.

AWARDS & RECOGNITION

CFO Recognition — BNSF Railway: for \$1M+ annual savings via ML-driven fraud detection system
Insta Award — Infosys: top-tier peer recognition for ML engineering impact and legacy system modernization